

IOAI Philippines 2026 - National Finals - Theory & Technical Concepts

- **Duration:** 1 Hour
 - **Total Items:** 30
 - **Instructions:** Select the single best answer for each question.
-

Section 1: Mathematics & Classical Machine Learning

Calculus, Linear Algebra, Optimization, and Statistical Foundations

1. [Feature Maps] Linear Separability

You have a 2D dataset that cannot be separated by a straight line. Class 0 consists of points inside a circle of radius 2 centered at $(0, 0)$. Class 1 consists of points outside that circle. Which of the following feature transformations $\phi(x_1, x_2)$ makes this dataset **linearly separable**?

- A) $\phi = [x_1, x_2, x_1 + x_2]$
- B) $\phi = [x_1^2, x_2^2]$
- C) $\phi = [\sin(x_1), \cos(x_2)]$
- D) $\phi = [x_1, x_2, x_1 x_2]$

2. [Optimization] The Hessian Matrix

You are training a model and reach a point where the gradient is exactly zero ($\nabla L = 0$). You compute the Hessian Matrix (second derivatives) and find its eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = -3$.

Geometrically, what is this point?

- A) A Local Minimum.
- B) A Local Maximum.
- C) A Saddle Point.
- D) A Global Minimum.

3. [Regularization Geometry] L1 vs. L2

Why does **L1 Regularization (Lasso)** tend to produce sparse solutions (weights that are exactly zero), while **L2 (Ridge)** does not?

- A) The L1 derivative is undefined at 0, forcing the optimizer to stop there.
- B) The L1 unit ball is a "diamond" shape that touches the loss contours at the axes.
- C) L1 regularization creates a convex loss surface, whereas L2 is non-convex.
- D) L1 adds noise to the gradient, effectively pruning small weights.

4. [Statistics] MLE Bias

You use Maximum Likelihood Estimation (MLE) to estimate the variance σ^2 of a Gaussian distribution using the formula $\hat{\sigma}^2 = \frac{1}{N} \sum (x_i - \mu)^2$. Is this estimator **unbiased**?

- A) Yes, it converges to the true variance.

- B) No, it underestimates the variance (Biased).
- C) No, it overestimates the variance (Biased).
- D) It depends on the mean.

5. [SVM] The Margin

You have a 1D dataset. The **Positive Support Vector** is at $x = 6$ and the **Negative Support Vector** is at $x = 2$. If you train a hard-margin SVM, where is the decision boundary and what is the size of the margin?

- A) Boundary: 4, Margin: 2
- B) Boundary: 4, Margin: 4
- C) Boundary: 3, Margin: 1
- D) Boundary: 5, Margin: 1

6. [PCA] Eigenvalues

You perform PCA on a dataset X of shape $(100, 2)$. The covariance matrix Σ has eigenvalues $\lambda_1 = 50$ and $\lambda_2 = 0.001$. What does this tell you about the data?

- A) The data is uniformly distributed in a circle.
- B) The data is essentially a straight line; it can be compressed to 1D with almost no loss.
- C) The data is 100-dimensional but projected down to 2D.
- D) The feature x_1 is 500x more important than x_2 .

7. [Gradient Descent] Update Math

Given loss function $L(w) = w^2$. Current weight $w = 3$. Learning rate $= 0.1$. What is the value of w after **one** gradient descent step?

- A) 2.4
- B) 2.7
- C) 3.6
- D) 0.6

8. [Naive Bayes] "Naive" Assumption

What is the specific mathematical assumption that makes Naive Bayes "Naive"?

- A) It assumes the prior probabilities are uniform.
- B) It assumes all features are conditionally independent given the class label.
- C) It assumes the data follows a Gaussian distribution.
- D) It assumes the classes are linearly separable.

9. [K-Means] Convergence

You run K-Means with $k = 3$. Which of the following is guaranteed to decrease (or stay the same) at every iteration of the algorithm?

- A) The silhouette score.
- B) The total "Within-Cluster Sum of Squares" (Inertia).
- C) The distance between centroids.
- D) The accuracy relative to ground truth labels.

10. [Decision Trees] Information Gain

You are splitting a node that contains 10 Positives and 10 Negatives. Which split gives the highest Information Gain?

- A) Left: [9 Pos, 1 Neg], Right: [1 Pos, 9 Neg].
 - B) Left: [5 Pos, 5 Neg], Right: [5 Pos, 5 Neg].
 - C) Left: [10 Pos, 0 Neg], Right: [0 Pos, 10 Neg].
 - D) Left: [6 Pos, 4 Neg], Right: [4 Pos, 6 Neg].
-

Section 2: Deep Learning Mechanics

Neural Networks, CNNs, Transformers, and Training Dynamics

11. [Symmetry Breaking] Initialization

You initialize a 2-layer neural network. You set all weights = 0 and all biases = 0. You use ReLU activation. You run one forward pass and one backward pass. What happens to the weights?

- A) They update normally.
- B) They remain exactly zero (Dead Network).
- C) They update, but every neuron in the layer learns the exact same feature.
- D) The biases update, but the weights do not.

12. [CNN Arithmetic] Output Volume

Input: $32 \times 32 \times 3$ image. Layer: 10 Filters of size 5×5 , Stride 1, Padding 0. What is the output volume size?

- A) $28 \times 28 \times 10$
- B) $32 \times 32 \times 10$
- C) $28 \times 28 \times 3$
- D) $14 \times 14 \times 10$

13. [Parameters] 1x1 Convolutions

You have a feature map of shape $64 \times 64 \times 256$. You apply a 1×1 **Convolution** with 64 filters. How many parameters does this layer have (excluding bias)?

- A) $64 \times 64 \times 64$
- B) $1 \times 1 \times 256 \times 64$
- C) $1 \times 1 \times 64$
- D) 256×64

14. [Receptive Field] Stacking Layers

You stack two 3×3 convolutional layers (Stride 1). What is the effective receptive field of the final output pixel relative to the input?

- A) 3×3
- B) 5×5
- C) 6×6
- D) $\times$

15. [Backprop] Vanishing Gradients

Why did researchers switch from **Sigmoid** activation to **ReLU** for deep networks?

- A) ReLU is differentiable everywhere.
- B) Sigmoid derivatives are always < 0.25 , causing gradients to vanish as they multiply through many layers.
- C) Sigmoid is not zero-centered.
- D) ReLU prevents the exploding gradient problem.

16. [Batch Normalization] Train vs. Test

How does Batch Normalization behave differently during **Inference (Testing)** compared to Training?

- A) It uses the batch statistics (mean/variance) of the current test batch.
- B) It uses the fixed "Running Mean" and "Running Variance" accumulated during training.
- C) It turns off completely.
- D) It normalizes by the instance mean (Instance Norm).

17. [Architectures] ResNet Skip Connection

Mathematically, what does the ResNet block $y = (x) + x$ allow the gradient to do during backpropagation?

- A) It forces the gradient to go through (x) .
- B) It allows the gradient to bypass (x) and flow directly to the previous layer.
- C) It squares the gradient, preventing it from vanishing.
- D) It acts as a regularizer, forcing weights to be small.

18. [Complexity] Self-Attention Memory

If you double the input sequence length from $N = 512$ to $N = 1024$ in a Transformer, how does the memory required for the **Attention Matrix** change?

- A) It doubles $(2x)$.
- B) It quadruples $(4x)$.
- C) It stays the same.
- D) It increases by N .

19. [Regularization] Dropout

You train with Dropout ($p = 0.5$). If a neuron's output is x during training, what should you do to the weights during **Inference** to maintain the same expected magnitude? (Assuming standard implementation).

- A) Multiply weights by 0.5.
- B) Multiply weights by 2.
- C) Do nothing.
- D) Set weights to 0.

20. [Pooling] Invariance

What specific property does **Max Pooling** introduce to a CNN?

- A) Rotational Invariance.
- B) Small Translation Invariance.

- C) Color Invariance.
 - D) Scale Invariance.
-

Section 3: Generative AI & Safety

LLMs, Diffusion Models, Sampling, and Ethics

21. [Sampling Math] Temperature Scaling

You have an LLM outputting raw logits $= [1.0, 3.0]$ for two tokens ("Cat", "Dog"). You apply **Temperature Sampling** with $T = 0.5$. What happens to the probability distribution after Softmax?

- A) The probabilities become closer to uniform (50/50).
- B) The probability of "Dog" (the higher logit) increases significantly, making the model more confident.
- C) The probability of "Dog" decreases.
- D) The order flips; "Cat" becomes the most likely token.

22. [Transformer Mechanics] Positional Encodings

Why must a Transformer add "Positional Encodings" to the input embeddings?

- A) Because the Self-Attention mechanism is **permutation invariant** (it treats "The dog bit the man" and "The man bit the dog" identically without them).
- B) Because the embeddings are too small to capture meaning.
- C) Because the gradients vanish without them.
- D) To prevent the model from overfitting to specific sentence lengths.

23. [LLM Training] Instruction Tuning

Pre-training an LLM on the "Pile" teaches it to predict the next token. What specific training phase teaches it to follow user commands like "Summarize this text"?

- A) Reinforcement Learning from Human Feedback (RLHF).
- B) Supervised Fine-Tuning (SFT) / Instruction Tuning.
- C) Contrastive Learning.
- D) Masked Language Modeling.

24. [RLHF] Reward Hacking

In RLHF, if you optimize the LLM against a learned Reward Model for too many steps without regularization, the LLM starts outputting nonsense that scores highly. This is an example of:

- A) Mode Collapse.
- B) Goodhart's Law (Reward Hacking).
- C) The Vanishing Gradient Problem.
- D) Catastrophic Forgetting.

25. [Adversarial Attacks] Gradient Method

A "White Box" adversarial attack (like FGSM) tricks an image classifier into seeing a Panda as a Gibbon. How is the noise generated?

- A) By randomizing pixels until the class changes.
- B) By calculating the gradient of the Loss with respect to the **Input Image** and stepping *up* the gradient (maximizing loss).
- C) By calculating the gradient of the Loss with respect to the **Weights** and stepping *down*.
- D) By inverting the softmax layer.

26. [Diffusion Models] The Forward Process

In a Denoising Diffusion Probabilistic Model (DDPM), what is the mathematical limit of the **Forward Process** () as time $t \rightarrow 0$?

- A) The image becomes a perfect reconstruction of the input.
- B) The image becomes a pure Isotropic Gaussian Noise distribution $(0, I)$.
- C) The image becomes a latent vector of dimension d .
- D) The image becomes zero (black).

27. [Tokenization] Byte Pair Encoding (BPE)

How does the BPE algorithm decide which characters to merge into a single token (e.g., merging 't' and 'h' into 'th')?

- A) It uses a pre-trained neural network to find semantic meaning.
- B) It merges the pair of tokens that appears most **frequently** in the corpus.
- C) It merges tokens randomly to increase diversity.
- D) It merges tokens based on English grammar rules.

28. [RAG] Retrieval Augmented Generation

You are building a RAG system. The "Retriever" finds relevant documents. What does the "Generator" (LLM) do with them?

- A) It retrains its weights on the documents for one epoch.
- B) It takes the documents as input context (in the prompt) and answers the user's question.
- C) It calculates the cosine similarity between the documents and the weights.
- D) It compresses the documents into a vector database.

29. [Fairness] Equalized Odds

- **Metric A:** The model approves loans for 50% of Men and 50% of Women.
- **Metric B:** The model approves loans for 80% of *qualified* Men and 80% of *qualified* Women.

Which metric corresponds to **Equalized Odds**?

- A) Metric A.
- B) Metric B.
- C) Both.
- D) Neither.

30. [RNN vs Transformer] Parallelism

Why can Transformers be trained much faster than Recurrent Neural Networks (RNNs/LSTMs) on massive datasets?

- A) Transformers have fewer parameters.
- B) RNNs must process tokens sequentially (step t depends on step $t - 1$), whereas Transformers

process the entire sequence in **parallel**.

C) Transformers do not use Backpropagation.

D) RNNs cannot use GPUs.